

Cross Asset Systematic Highlights

Ducks in a row: Improving our Positioning Strategy with Crowding metrics

- Unlike the murky world of positioning *indicators*, we have for a long time offered a *clear* methodology for a CFTC Positioning *Strategy*, which is our '[Variation in Hedging Pressure](#)' (VHP).
- We improved on [Positioning by adding Momentum](#), since we don't just want things that are over- or under-owned, we want things where Momentum agrees, and the Positioning is starting to revert. What about Crowding, and how does it fit into Positioning?
- The current positioning strategy traded the front month, and the CFTC data is *aggregated*, making it difficult to draw conclusions about the forward months. So we explore ways to assess most and least crowded contracts. Our Crowding indicator here is [FOMO](#), which is the z-score of the recent Sharpe ratio relative to the long-run Sharpe.
- Both Momentum & Crowding produce similar improvements to the Positioning strategy in isolation (.38 to ~.5), but the combination of all three produces a Sharpe of 0.86.
- Where are all three signals firing together currently? Bullish on Heating Oil and Gold, with the FOMO crowding identifying the favorable contracts as HO1 and GC4, respectively.
- Why are positioning strategies special? They have often done well when other risk premia are failing, so we expect diversification improvements from adding 'Position+Momentum+Crowding' to portfolios, whether risk premia or the 60/40 portfolio.
- This exercise focuses on commodities, and we may explore similar approaches in other classes.

Cross Asset Systematic Strategy

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Figure 1: Focusing on Commodities, the Positioning-only approach has had a long history as a durable alpha source. Adding both momentum and FOMO crowding produces large improvement, with FOMO identifying best forward contracts to trade.



Figure 2: We have applied the FOMO crowding idea across the asset classes. For S&P500, we are at ~+1 z-score, below our +1.5 threshold. Stay long stocks.

Average z-score of 3m, 6m, and 12m Sharpe relative to long-run history



See page 10 for analyst certification and important disclosures.

Market Views

The chart below really brings home the message on stocks, as well as the historical experience post-GFC. In recent decades, we've had a market driven by multiple expansion as earnings expectations drifted down for much of the period. Fast forward to the right side of the graph, and we see the impact the geopolitical stress has had on multiples, and even with the recent bounce, multiples are well below highs. Even with the still depressed multiples, exceptional results on the earnings side justify the lofty US index levels. With 88% of S&P500 having reported Q1 earnings, 82% have beat EPS estimates (vs a long-term average of ~55%), with 10 out of 11 sectors up on a yoy basis. So we've had a nice broadening of the earnings growth, justifying a further catch-up from Cyclical Value sectors still not back at highs. The rising bond yields (US 30y yield at 5.1%) and supercore inflation inflecting up also reflect the idea that bond yields will continue upward to challenge stocks, although the view from Risk Premia by Pairs was that stocks have more room to go for now. Of course, if we zoomed in on the recent stock index moves, they look fairly extreme, but alongside the improving earnings, the moves look reasonable.

Given the recent strong jobs print, consider what it means for stock vs bond risk premia comparisons. When were investors most tolerant of expensive valuations and tight risk premia? When they were expecting something good not yet expressed in the numbers, e.g., productivity boost from Dotcom or AI. In the immediate aftermath of Iran, there were more persistent growth worries in relation to an already sticky inflation picture. So the recent strong jobs print confirms the strength of the economy; with more growth than inflation, stocks can tolerate still tighter stock vs bond risk premia. Given the pressure on both US and Iran, a peace deal would release a whoosh of pent-up supply of oil, and given oil vs US 10y yield correlation at extreme highs, would bring down bond yields. This would in turn widen the stock vs bond risk premia while bringing vols down, justifying the next leg of the stock rally. However, apart from the usual stock vs bond risk premia rationale to underweight bonds, there would be additional reason to underweight bonds at that point. As discussed [earlier](#), the uncertainty in the Middle East will keep oil prices from getting back to their old low levels, not to mention oil importers building reserves. So the celebratory moment of a peace deal offers an opportunity to get short duration at a good level as the post-war world will be more complicated and expensive, feeding stickier inflation, with tanker trips to and from the Middle East becoming more costly.

Figure 3: Who's driving? Multiples or Earnings? For most of post-GFC, it was multiples, but in recent period earnings have more than made up the multiples, reflecting risk aversion.

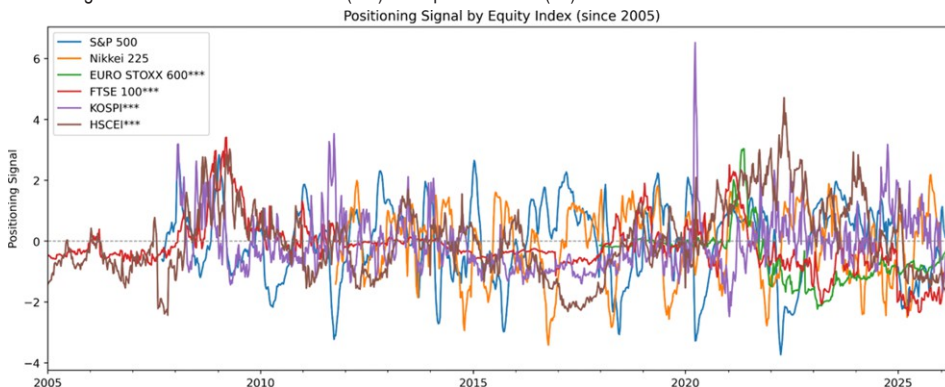


Source: J.P. Morgan Research, Bloomberg Financial L.P.

We fully acknowledge the positioning is high for both S&P500 and Nikkei, using similar [Variation in Hedging Pressure](#) measures, although this is well off the highs, indicating how stretched positioning could get with more conviction. Meanwhile, for crowding, our FOMO crowding measure calculates the z-score of recent Sharpe ratios vs. the long-term history. S&P500 is somewhat stretched at a +1 z-score, and generally our cutoff would be +1.5 z-score. Bear in mind that the Sharpe ratios took a hit during the stock sell-off. So this, coupled as well with the strong positive momentum, suggests it is premature to call a top in equities.

Figure 4: Our positioning signals across indices show positioning highest in S&P500 and Nikkei, but well off the highs. For the ROW, positioning not a factor. With momentum positive and positioning not at highs, it would seem premature to call a market top.

Positioning indicators from futures markets (no*) and option markets (***)



Source: J.P. Morgan Research, Bloomberg Financial L.P.

Figure 5: For S&P500, the last time our FOMO crowding indicator saw an extreme signal (>1.5) was ahead of the July-August 2024 stock pullback. Not currently an issue.

Average z-score of 3m, 6m, and 12m Sharpe relative to long-run history



Source: J.P. Morgan Research, Bloomberg Financial L.P.

Introduction to Our Back-Test

Cross-sectional positioning strategies based on CFTC Commitments of Traders (COT) data have long been recognized as a robust source of systematic alpha in commodity futures markets. In [Profiting from Positioning Dynamics in Commodity Futures](#), we constructed a “Variation in Hedging Pressure” factor that captured positioning dynamics of Hedgers and Speculators and demonstrated its robust return-generating capabilities in commodity and FX futures. In a later [research piece](#), we followed similar ideas and designed a standardized indicator tracking the spread between short-term and long-term

averages of hedging pressure, extending the approach to the cross-asset space covering equities and bonds as well.

While the baseline positioning strategy delivers attractive stand-alone performance, two natural extensions warrant investigation. First, raw positioning signals can occasionally fire prematurely – a crowded position may remain crowded for extended periods before reversing. To mitigate this, we explored the addition of a momentum overlay as a confirmation filter in [Combining Positioning Dynamics and Price Momentum in Cross Asset Futures](#) and [Tactical Trifecta](#). The latest trade recommendations and strategy performance can be found in our PCA Dislocation, Positioning and Momentum Dashboard. Second, the choice of which contract along the futures curve to trade has historically been made by default – the front contract – without explicit consideration of relative value across the term structure. Yet positioning, carry, and speculative crowding can vary meaningfully between the 1st, 2nd, 3rd, and 4th generic contracts of the same underlying commodity, creating opportunities for more efficient contract selection. In this paper, we use our crowding metric [FOMO](#) to refine at which point along the futures curve to express the positioning view.

We focus on 20 liquid commodities traded on US exchanges and start by analyzing the improvements of momentum and FOMO on the baseline positioning strategy separately. Finally, we combine these two extensions into a unified framework that uses positioning for asset selection, FOMO for contract selection, and momentum for trade confirmation.

Trading Front Contracts with CFTC Positioning Signals

Recall we defined the hedging pressure as the net long positions of the legacy non-commercial category (NCN) scaled by open interest for commodity futures:

$HP_{i,t} = \frac{NCN_{i,t}}{OI_{i,t}}$ where $OI_{i,t}$ denotes the open interest of future i in week t . We then generate the exponentially weighted moving averages of the hedging pressure over a shorter span (five weeks) and a longer span (from half-year to five years) and compute the spread of the short-term and long-term averages. The spread is then normalized over the past 52 weeks. Taking the one-year case as an example, the hedging pressure score becomes:

$$HP_Spread_{i,t} = EWMA_{HP_{i,t}}(\text{span} = 5 \text{ weeks}) - EWMA_{HP_{i,t}}(\text{span} = 52 \text{ weeks})$$

$$HP_Score_{i,t} = \frac{HP_Spread_{i,t} - \overline{HP_Spread_i}}{\sigma_{HP_Spread_i}}$$

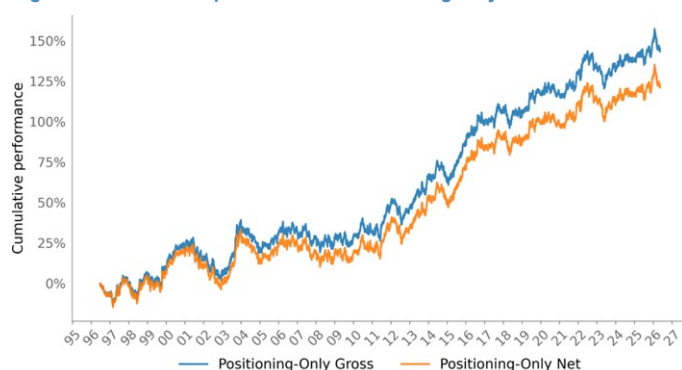
where $\overline{HP_Spread_i}$ and $\sigma_{HP_Spread_i}$ are sample mean and standard deviation of the spread over the previous 52 weeks.

To enhance the robustness of our indicator, we generate the longer-term averages over half, one, two, three, and five years separately. The final indicator is the average of the scores across these five horizons. When the score is high (low), the spread of hedging pressure is increasing (decreasing), meaning hedgers are currently hedging more (less) compared to their long-term mean, leading to negative (positive) forward returns under our assumption.

The back-test is implemented as a cross-sectional long-short portfolio rebalanced weekly. At each rebalance, we rank the universe of 20 commodities by their latest CFTC positioning signals, shorting the top 30% (most over-owned) and going long the bottom 30% (most under-owned). The magnitude of the weights is inversely proportional to the realised volatility based on a rolling 1Y covariance matrix, and the overall portfolio is scaled to a target annualized volatility of 10%.

We trade the front-month (1st generic) contracts exclusively in this baseline implementation. Below we present both the gross and net performance of the CFTC positioning strategy, taking into account the realistic transaction and running costs.

Figure 6: Cumulative performance: Positioning only



Source: J.P. Morgan Research, Bloomberg Financial L.P.

Table 1: Performance statistics

	Start	End	Ann. Return	Ann. Vol	Sharpe	Max DD
Positioning-Only Gross	1996-06-24	2026-05-14	4.65%	10.47%	0.44	-24.11%
Positioning-Only Net	1996-06-24	2026-05-14	3.93%	10.47%	0.38	-24.96%

Source: J.P. Morgan Research, Bloomberg Financial L.P.

The baseline strategy delivers a robust performance over the full back-test period with a Sharpe of 0.38, confirming the persistent return-generating capacity of CFTC-based positioning signals in commodity futures.

Enhancing Positioning Signals with Momentum Filters

As stated in the introduction, while CFTC positioning captures the crowding dimension of a market, it does not directly observe whether prices are already moving in the direction the signal anticipates. To address this, we add a momentum signal as a confirmation filter and only trade the markets where both signals align. Specifically, we employ the delta-of-the-straddle signal across look-back windows of 32 days, 3 months, 6 months, 1 year, and 2 years as described in [Designing a Robust Trend Following System](#).

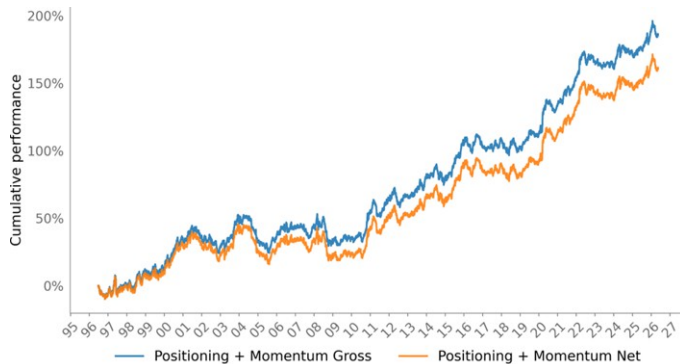
At each weekly rebalance, we first identify the most over-owned (top 30%) and most under-owned (bottom 30%) assets as before. We then apply the following filtering rule:

- A long position (under-owned asset) is only retained if its momentum is positive, indicating that prices have already started to move up in line with the contrarian positioning view.
- A short position (over-owned asset) is only retained if its momentum is negative, indicating that prices have already started to move down.

Assets that fail the momentum check are dropped from the portfolio at that rebalance. The remaining names are sized using inverse-volatility weighting based on a rolling 1Y covariance matrix, with the overall portfolio scaled to an annualized volatility target of 10%. As in the baseline, we trade the front-month (1st generic) contracts exclusively,

and we report both gross and net performance below.

Figure 7: Cumulative performance: Position + Momentum



Source: J.P. Morgan Research, Bloomberg Financial L.P.

Table 2: Performance statistics

	Start	End	Ann. Return	Ann. Vol	Sharpe	Max DD
Positioning + Momentum Gross	1996-06-24	2026-05-14	6.00%	10.18%	0.59	-25.33%
Positioning + Momentum Net	1996-06-24	2026-05-14	5.19%	10.19%	0.51	-26.31%

Source: J.P. Morgan Research, Bloomberg Financial L.P.

The combination of positioning and momentum boosts the Sharpe ratio to 0.51 after cost. This is consistent with our findings in [Combining Positioning Dynamics and Price Momentum in Cross Asset Futures](#) and [Tactical Trifecta](#).

Refining Positioning Signals with Cross-Contract FOMO Analysis

CFTC data aggregates open interest and trader positions across all contract months for a given commodity, rather than breaking them down by individual delivery months or generic contract rankings (e.g., 1st, 2nd, 3rd contract). While this provides a robust asset-level view of speculative crowding, it does not differentiate positioning along the futures curve of the same underlying commodity. Consequently, the positioning signal alone cannot identify where on the curve speculators are most concentrated, leaving the choice of trading contract unaddressed.

To capture contract-level crowding, we employ our FOMO indicator, defined as the average z-score (using an expanding window) of the 3-, 6-, and 12-month Sharpe ratios for each asset. It measures how stretched the near-term Sharpe ratio is relative to its long-term history. We compute FOMO independently for each generic contract (1st, 2nd, 3rd, 4th), so the same underlying commodity can exhibit very different FOMO readings across the curve.

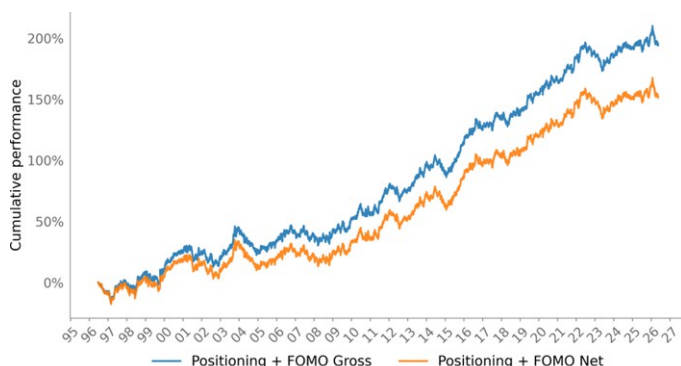
Combining the two, we keep CFTC positioning as the primary asset-selection signal and use cross-contract FOMO purely as a tool for choosing the optimal expression of the trade along the futures curve. Importantly, the FOMO selection relies only on the cross-contract ranking, not the absolute level – the chosen contract simply needs to have the most extreme FOMO among the 1st–4th generics for that commodity, even if the absolute reading is not directionally aligned. Specifically, at each weekly rebalance the strategy proceeds in two stages:

1. **Asset selection (CFTC positioning):** Rank the universe of 20 commodities by their HP_Score and identify the top 30% (most over-owned, candidates for shorting) and bottom 30% (most under-owned, candidates for longing).
2. **Contract selection (FOMO):** For each selected commodity, compute FOMO across the 1st, 2nd, 3rd, and 4th generic contracts and:

- a. For **long** candidates (under-owned), pick the contract with the **lowest** FOMO
- b. For **short** candidates (over-owned), pick the contract with the **highest** FOMO

The selected contracts are sized using inverse-volatility weighting based on a rolling 1Y covariance matrix, and the overall portfolio is scaled to an annualized volatility target of 10%. Unlike the previous sections, the trade is no longer constrained to the front-month contract – the FOMO ranking can select any of the first four generics, though only one contract will be traded per asset. Below we present both the gross and net performance of the combined CFTC positioning + FOMO strategy.

Figure 8: Cumulative performance: Positioning + FOMO Crowding



Source: J.P. Morgan Research, Bloomberg Financial L.P.

Table 3: Performance statistics

	Start	End	Ann. Return	Ann. Vol	Sharpe	Max DD
Positioning + FOMO Gross	1996-06-24	2026-05-14	6.28%	10.64%	0.59	-21.45%
Positioning + FOMO Net	1996-06-24	2026-05-14	4.90%	10.65%	0.46	-22.50%

Source: J.P. Morgan Research, Bloomberg Financial L.P.

Compared with the baseline of trading only the 1st generic contracts according to CFTC positioning, adding FOMO and expanding the contract set improves the net performance Sharpe ratio from 0.38 to 0.46.

To illustrate how the strategy translates into actionable trades in real time, the table below presents the latest snapshot of FOMO scores across the 1st–4th generic contracts for each commodity, with the contract selected by the strategy highlighted: green for long positions and red for shorts.

Figure 9: Latest FOMO across generic contracts (as of 2026-05-14)

Green: long positions; Red: short positions - selected by the Positioning + FOMO strategy

	Name	1st	2nd	3rd	4th
CL	WTI Crude Oil	1.17	1.31	1.32	1.32
HO	Heating Oil	1.39	1.61	1.63	1.65
XB	Gasoline	2.00	1.87	1.83	1.87
NG	Natural Gas	0.26	-0.34	-0.59	-0.80
GC	Gold	0.32	0.31	0.21	0.18
SI	Silver	1.12	1.21	1.12	1.08
HG	Copper	0.79	0.78	0.77	0.74
C	Corn	0.17	0.30	0.43	0.54
S	Soybean	0.08	0.26	0.38	0.59
BO	Soybean Oil	1.84	1.83	1.79	1.77
CT	Cotton	1.50	1.68	1.85	1.88
KC	Coffee	-0.56	-0.48	-0.58	-0.63
SB	Sugar	0.07	0.19	0.21	0.28
W	Wheat-SRW	1.12	1.04	0.99	1.02
LC	Live Cattle	0.89	0.94	0.90	0.72
LH	Lean Hogs	-0.26	0.06	0.09	0.33
KW	Wheat-HRW	1.19	1.18	1.13	1.11
SM	Soybean Meal	0.21	0.04	-0.14	-0.26
PL	Platinum	0.66	0.79	0.26	0.69
CC	Cocoa	-0.50	-0.40	-0.41	-0.41

Source: J.P. Morgan Research, Bloomberg Financial L.P.

From the table, we observe that cross-contract dispersion is non-trivial and often pushes the trade away from the front contract. Several selected positions sit at the back of the curve rather than the default front month, for example, NG4, C4, CT4, GC4, and KC4. FOMO spreads from front to back exceed **1.0 standard deviation** in Natural Gas (1.06) and reach **0.4–0.6 std** in Lean Hogs (0.59), Platinum (0.53), Soybeans (0.51), Soybean Meal (0.47). This is clear evidence that the term structure carries information not captured by aggregate CFTC positioning.

Integrating Momentum Filters with FOMO-Based Selection

Having established that (i) the momentum overlay improves the front-contract positioning strategy and (ii) cross-contract FOMO selection improves the unfiltered positioning strategy, the natural next step is to combine all three signals into a single unified framework. The goal is to retain the directional edge of CFTC positioning, the curve-aware contract selection of FOMO, and the entry-timing benefit of momentum confirmation.

The strategy proceeds at each weekly rebalance in three stages:

1. **Asset selection (CFTC positioning):** Rank the universe of 20 commodities by their HP_Score. The top 30% (most over-owned) are flagged as short candidates and the bottom 30% (most under-owned) as long candidates.
2. **Contract selection (FOMO):** For each flagged commodity, compute FOMO across the 1st, 2nd, 3rd, and 4th generic contracts. For long candidates pick the contract with the lowest FOMO; for short candidates pick the contract with the highest FOMO.
3. **Trade confirmation (momentum):** Apply the delta-of-the-straddle momentum filter to the selected contract. A long position is retained only if its momentum signal is positive; a short position is retained only if its momentum signal is negative. Candidates that fail the momentum check are dropped from the portfolio at that rebalance.

The surviving names are sized using inverse-volatility weighting based on a rolling 1Y covariance matrix, and the overall portfolio is scaled to an annualized volatility target of 10%. As in the CFTC positioning + FOMO section, the trade is no longer constrained to the front-month contract –the FOMO ranking can pick any of the first four generics, though only one contract is traded per asset. Below we present both the gross and net performance of the combined CFTC positioning + FOMO + momentum strategy.

Figure 10: Cumulative performance: Pos + Mom + FOMO



Source: J.P. Morgan Research, Bloomberg Financial L.P.

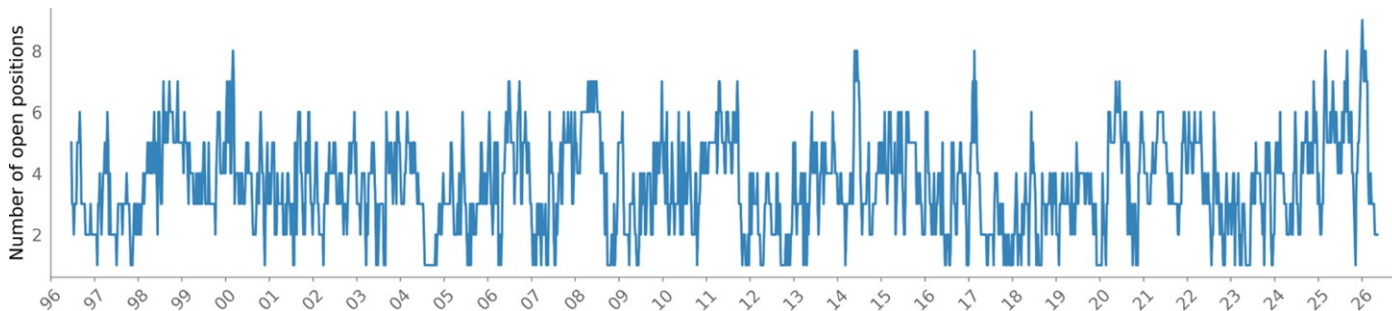
Table 4: Performance statistics

	Start	End	Ann. Return	Ann. Vol	Sharpe	Max DD
Positioning + FOMO + Momentum Gross	1996-06-24	2026-05-14	9.77%	10.30%	0.95	-19.74%
Positioning + FOMO + Momentum Net	1996-06-24	2026-05-14	8.83%	10.30%	0.86	-20.40%

Source: J.P. Morgan Research, Bloomberg Financial L.P.

The full three-signal strategy lifts the net Sharpe ratio to 0.86, compared with 0.46 for positioning + FOMO and 0.38 for the baseline. The momentum filter naturally reduces the average number of positions held at any given time, but the surviving trades benefit from the directional confirmation provided by price action. So, on average, we have 3.6 positions per week satisfying all three criteria.

Figure 11: Number of open positions of the Positioning + FOMO + Momentum strategy



Source: J.P. Morgan Research, Bloomberg Financial L.P.

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